

Extended Physics-Informed Neural Networks (XPINNs): A Generalized Space-Time Domain Decomposition Based Deep Learning Framework for Nonlinear Partial Differential Equations

Ameya D. Jagtap^{1,*} and George Em Karniadakis^{1,2}

¹ Division of Applied Mathematics, Brown University, 182 George Street,
Providence, RI 02912, USA.

² Pacific Northwest National Laboratory, Richland, WA 99354, USA.

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Abstract. We propose a generalized space-time domain decomposition approach for the physics-informed neural networks (PINNs) to solve nonlinear partial differential equations (PDEs) on arbitrary complex-geometry domains. The proposed framework, named eXtended PINNs (XPINNs), further pushes the boundaries of both PINNs as well as conservative PINNs (cPINNs), which is a recently proposed domain decomposition approach in the PINN framework tailored to conservation laws. Compared to PINN, the XPINN method has large representation and parallelization capacity due to the inherent property of deployment of multiple neural networks in the smaller subdomains. Unlike cPINN, XPINN can be extended to any type of PDEs. Moreover, the domain can be decomposed in any arbitrary way (in space and time), which is not possible in cPINN. Thus, XPINN offers both space and time parallelization, thereby reducing the training cost more effectively. In each subdomain, a separate neural network is employed with optimally selected hyperparameters, e.g., depth/width of the network, number and location of residual points, activation function, optimization method, etc. A deep network can be employed in a subdomain with complex solution, whereas a shallow neural network can be used in a subdomain with relatively simple and smooth solutions. We demonstrate the versatility of XPINN by solving both forward and inverse PDE problems, ranging from one-dimensional to three-dimensional problems, from time-dependent to time-independent problems, and from continuous to discontinuous problems, which clearly shows that the XPINN method is promising in many practical problems. The proposed XPINN method is the generalization of PINN and cPINN methods, both in terms of applicability as well as domain decomposition approach, which efficiently lends itself to parallelized computation. The XPINN code is available on <https://github.com/AmeyaJagtap/XPINNs>.

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*Corresponding author. Email addresses: ameyadjagtap@gmail.com, ameya_jagtap@brown.edu (A. D. Jagtap), george_karniadakis@brown.edu (G. E. Karniadakis)